



Ishan Arora

Bachelor of Technology
in Metallurgical and Materials Engineering
Indian Institute Of Technology, Ropar

+91-8595626909
aroraishan175@gmail.com
GitHub
LinkedIn

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.91	2024-2028
Senior Secondary	Central Board of Secondary Education	91.8%	2024
Secondary	Central Board of Secondary Education	96.2%	2022

PROJECTS

- Sovereign Arbitrage & Gold Valuation Engine**
Python (scikit-learn, pandas), Machine Learning, Macroeconomic Data Architecture [Github](#)
 - Engineered a predictive macroeconomic valuation engine using a regularized Ridge Regression ML pipeline to forecast gold's fundamental baseline price against historic 2005–2022 market regimes.
 - Applied L2 regularization algorithms to manage multi-collinearity across 5 core time-series features (CPI, 10-Year Real Yields, Bond Yields, DXY, WTI Crude) with a training R^2 of 0.9015.
 - Projected an out-of-sample baseline showing that under historical macro relationships gold would not exceed \$1,750/oz today; achieved a test R^2 of -9.4613 to mathematically prove structural decoupling driven by geopolitical factors.
- Credit Card Fraud Detection and Risk Pipeline**
Python, XGBoost, Scikit-Learn, Pandas [Github](#)
 - Engineered an end-to-end machine learning pipeline to detect anomalous transactions in a highly imbalanced dataset (99.8% normal data). Developed an XGBoost ensemble model that achieved ~80% recall and ~90% precision, optimizing risk assessment and minimizing simulated financial losses compared to baseline linear models.
- Algorithmic Trading Signal Predictor**
Python (yfinance, pandas), Natural Language Processing [Github](#)
 - Engineered an automated trading pipeline that predicts Apple (AAPL) stock trends by merging 3 years of historical market data with financial news sentiment.
 - Processed 1,500+ financial headlines to quantify daily market sentiment, training a predictive model that achieved 93% out-of-sample accuracy while eliminating volume-based overfitting.
- Empirical IS-LM Modeling & 2026 Macro-Fiscal Simulation**
Python (pandas, statsmodels), OLS Regression, Macroeconomic [Github](#)
 - Built a Python pipeline using pandas and statsmodels to ingest, wrangle, and preprocess raw RBI datasets into a clean, unified quarterly macroeconomic model.
 - Applied OLS regression to mathematically derive the empirical slopes of India's IS and LM curves, executing a simultaneous equation solver to simulate a 10% fiscal expansion shock and quantify the downstream basis-point impacts on sovereign bond yields, currency flows, and equity valuations.

TECHNICAL SKILLS

- Languages & Databases:** Python, SQL, MySQL, DBMS
- Libraries:** Numpy, Pandas, Matplotlib, Seaborn, Sklearn, yfinance, Hugging Face, PyTorch, FastAPI
- Tools & Software:** Linux, Git, GitHub, Jupyter, VS Code, Excel, Power BI, Adobe Photoshop, Canva
- Soft Skills:** Problem Solving, Analytical Thinking, Risk Assessment, Professional Communication, Team Collaboration

KEY COURSES TAKEN

- CSE & Maths:** Differential Equations, Probability and Statistics, Linear Algebra, Calculus
- Materials:** Material Modelling Using Python, Material Characterization, Machine Learning in Materials Science
- Electrical and Electronics:** Introduction to Electrical Engineering, Basic Electronics

POSITIONS OF RESPONSIBILITY

- Assistant Internship Coordinator (Non Core),**CDPC *Apr. 2026 - Present*
- ISMP Mentor,**ISMP *Jul. 2025 - Present*
- Hospitality Head,**Finance and Economics Summit *Mar. 2026 - May 2026*
- Event Management Co-Head,**Zeitgeist *Jan. 2026 - Feb. 2026*
- Graphic Designing Head,**Malhar *Sep. 2025 - Sep. 2026*
- Social Media Coordinator,**Panache (The Fashion Society), Metsoc *Jul. 2025 - Apr. 2026*

HONORS & EXTRACURRICULARS

- Runner-Up, FinQuiz,**Secured 2nd position among 50+ participants in the finance competition at IIT Ropar *2026*
- Runner-Up, Vaults & Valuations,**Secured 2nd rank in the Elumiere'26 academic simulation at SSCBS (Certificate) *2026*
- Inter IIT Cultural Meet 8.0,**Represented the institute in the flagship inter-college fashion event *2026*
- 3rd Position, Venture Velocity,**Pitched 'VyapaarNest', an MSME credit solution, at Kirori Mal College (Certificate) *2026*
- Gold Medalist, Table Tennis,**Secured 1st position in the Zonal level Table Tennis championship *2022*

CERTIFICATIONS

- Financial Markets,**Yale University [Link](#)
- Equity Derivatives (Basic),**NISM [Link](#)
- A1 German Certification,**Goethe-Institut [Link](#)